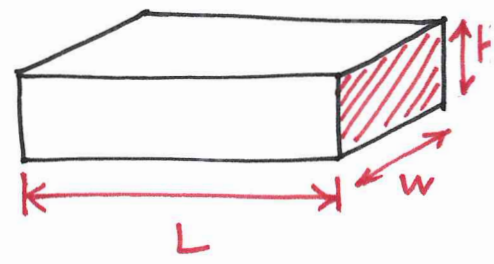


① Given the following microchannel find the pressure drop ΔP required to obtain a flow rate $Q = 1 \mu\text{e/s}$



a) with $L = 1 \text{ mm}$

b) with $L = 1 \text{ cm}$

DATA: $W = 100 \mu\text{m}$, $h = 10 \mu\text{m}$, $\eta_{\text{WATER}} = 10^{-3} \text{ Pa}\cdot\text{s}$

$$Q = \frac{\Delta P}{R}$$

$$\frac{W}{h} = 10 \gg 1.4 \Rightarrow R = \frac{12 \eta \cdot L}{(1 - 0.63 \frac{h}{W}) \cdot h^3 \cdot W}$$

a) $L = 1 \text{ mm}$

$$= \frac{12 \cdot 10^{-3} \cdot 10^{-3}}{(1 - 0.63 \cdot \frac{1}{10}) (10 \cdot 10^{-6})^3 \cdot 100 \cdot 10^{-6}} = 128 \cdot 10^{12} \left[\frac{\text{Pa}\cdot\text{s} \cdot \text{m}}{\text{m}^4} \right] = 128 \cdot 10^{12} \left[\frac{\text{Pa}\cdot\text{s}}{\text{m}^3} \right]$$

$$\Delta P = Q \cdot R = 1 \mu\text{e/s} \cdot 128 \cdot 10^{12} \cdot \frac{\text{Pa}\cdot\text{s}}{\text{m}^3}$$

$$1 \text{ m}^3 = 1000 \text{ l}$$

$$1 \mu\text{l} = \frac{1}{1000} \text{ m}^3 \cdot 10^{-6}$$

$$= 10^{-9} \cdot 128 \cdot 10^{12} \frac{\text{m}^3}{\text{s}} \cdot \frac{\text{Pa}\cdot\text{s}}{\text{m}^3} = \boxed{128 \cdot 10^3 \text{ Pa}} \quad [10^5 \text{ Pa} = 1 \text{ atm}]$$

1.2 atm

b) $L = 1 \text{ cm}$

$$R \text{ is linear with } L \Rightarrow R|_{1 \text{ cm}} = R|_{1 \text{ mm}} \cdot 10$$

$$\Delta P|_{1 \text{ cm}} = 1.2 \text{ atm} \cdot 10 = \boxed{12 \text{ atm}}$$

very high pressure !!!



above syringe max pressure.

$$14.7 \text{ PSI/atm}$$

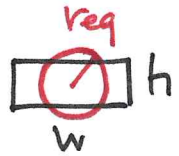
$$P|_{\text{max}} = 100 \text{ PSI}$$

$$12 \text{ atm} \cdot 14 \frac{\text{PSI}}{\text{atm}} = 176 \text{ PSI} > P_{\text{max}}$$

② For a rectangular microchannel, compare the estimate of the fluidic resistance

$$w = 10 \mu\text{m}$$

$$h = 10 \mu\text{m}$$



$$r_{eq} = \frac{2A}{P} \rightarrow$$

$$R_{eq} = \frac{8\eta L}{\pi r^4}$$

vs

$$R = \frac{12\eta L}{(1 - 0.63 \frac{h}{w}) h^3 w}$$

$$r_{eq} = \frac{2(w \cdot h)}{2(w+h)} = \frac{w \cdot h}{w+h}$$

$$r_{eq} = \frac{10 \cdot h \cdot h}{11h} = 0.9h$$

$$R_{eq} = \frac{8\eta L}{\pi \left(\frac{w \cdot h}{w+h} \right)^4}$$

$w = 10h$ $w \gg h$ thin channels

$$\sim \frac{8\eta L}{\pi \left(\frac{wh}{w} \right)^4} = \frac{8\eta L}{\pi h^4}$$

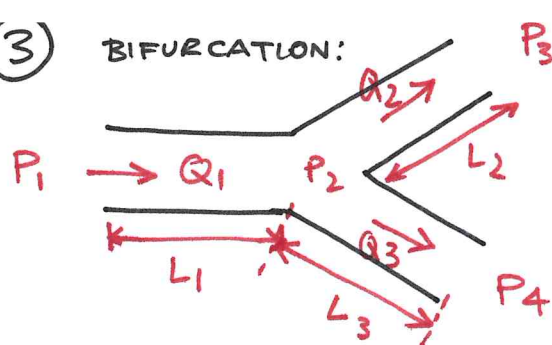
$$R = \frac{12\eta L}{(1-0)h^3 w} = \frac{12\eta L}{h^3 w} = \frac{12\eta L}{h^3 \cdot 10h} = \frac{12}{10} \frac{\eta L}{h^4}$$

$$\frac{8}{\pi} \text{ vs } 1.2$$

$$2.54 \text{ vs } 1.2$$

r_{eq} approx. estimates double value

(3) BIFURCATION:



Find the flow rates Q_1, Q_2, Q_3 given the pressures:

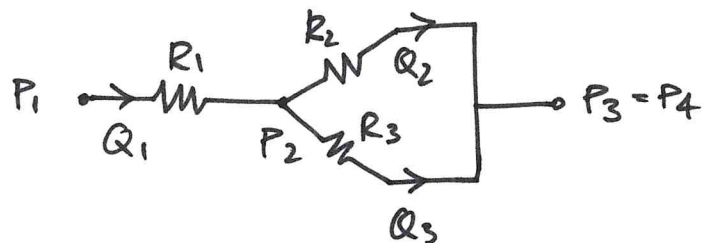
$$P_1 = 0.11 \text{ MPa} \quad L_1 = 0.5 \text{ cm}$$

$$P_3 = 0.1 \text{ MPa} \quad L_2 = 1 \text{ cm}$$

$$P_4 = 0.1 \text{ MPa} \quad L_3 = 2 \text{ cm}$$

Circular channel, radius $r = 20 \mu\text{m}$

Equivalent model:



IN WATER $\eta = 10^{-3} \text{ Pa}\cdot\text{s}$

Circular $R = \frac{8\eta \cdot L}{\pi r^4}$

$$R_1 = \frac{8\eta L}{\pi (20 \cdot 10^{-6})^4} = \frac{8 \cdot 10^{-3} \text{ Pa}\cdot\text{s} \cdot 0.5 \cdot 10^{-2} \text{ m}}{\pi (20 \cdot 10^{-6} \text{ m})^4} = 79.5 \cdot 10^{12} \frac{\text{Pa}\cdot\text{s}}{\text{m}^3}$$

$$R_2 = R_1 \cdot L_2 / L_1 = R_1 \cdot 2 = 159 \cdot 10^{12} \frac{\text{Pa}\cdot\text{s}}{\text{m}^3}$$

$$R_3 = R_1 \cdot L_3 / L_1 = R_1 \cdot 4 = 318 \cdot 10^{12} \frac{\text{Pa}\cdot\text{s}}{\text{m}^3}$$

$$Q_1 = \frac{\Delta P|_{\text{TOT}}}{R_{\text{TOT}}} = \frac{\Delta P|_{\text{TOT}}}{R_1 + R_2 // R_3} = \frac{110 \text{ kPa} - 100 \text{ kPa}}{(79.5 \cdot 10^{12} + 106 \cdot 10^{12}) \frac{\text{Pa}\cdot\text{s}}{\text{m}^3}} = 54 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}}$$

$$R_2 // R_3 = 106 \cdot 10^{12} \frac{\text{Pa}\cdot\text{s}}{\text{m}^3} \quad 54 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}} \rightarrow 54 \cdot 10^{-12} \cdot 10^9 \cdot 60 = 3 \mu\text{e}/\text{min}$$

$$P_2 = P_1 - \Delta P_1 = P_1 - Q_1 \cdot R_1 = 110 \text{ kPa} - 54 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}} \cdot 79.5 \cdot 10^{12} \frac{\text{Pa}\cdot\text{s}}{\text{m}^3} = 110 \text{ kPa} - 4.3 \text{ kPa} = 105.7 \text{ kPa}$$

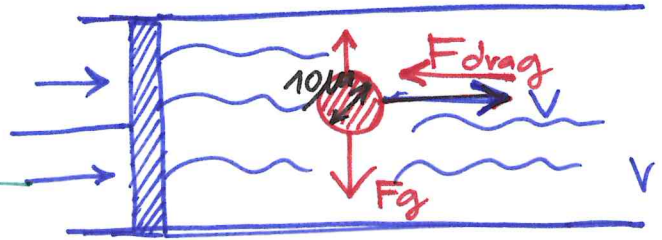
$$Q_2 = \frac{P_2 - P_3}{R_2} = \frac{105.7 \text{ kPa} - 100 \text{ kPa}}{159 \cdot 10^{12} \frac{\text{Pa}\cdot\text{s}}{\text{m}^3}} = 35.8 \cdot 10^{-12} \text{ m}^3/\text{s} \quad (2.15 \mu\text{e}/\text{min})$$

$$Q_3 = \frac{P_2 - P_4}{R_3} = \frac{105.7 \text{ kPa} - 100 \text{ kPa}}{318 \cdot 10^{12} \frac{\text{Pa}\cdot\text{s}}{\text{m}^3}} \sim \frac{Q_2}{2} = 18 \cdot 10^{-12} \text{ m}^3/\text{s} \quad (1 \mu\text{e}/\text{min})$$

(4) Compare gravity and drag forces for a $10\mu\text{m}$ particle
 a) of polystyrene and iron in water

$$r = 5\mu\text{m}$$

$$V = 1\mu\text{m/ms}$$



Drag is present if the particle is moved with respect to the fluid by an external force

$$F_{\text{drag}} = 6 \cdot \pi \cdot \eta \cdot r \cdot V = 6\pi \cdot 10^{-3} \text{Pa} \cdot \text{s} \cdot 5\mu\text{m} \cdot 1\mu\text{m/ms} = 0.1 \text{ nN}$$

$$F_g = (\rho_p - \rho_{\text{fluid}}) \cdot g \cdot \frac{4}{3} \pi r^3$$

density
of the
particle

density
of the
fluid

Archimedes's push

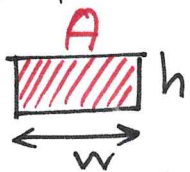
$$\rho_{\text{fluid}} = 1 \text{ g/cm}^3 (1000 \text{ kg/m}^3)$$

$$\rho_{\text{poly}} \sim 1 \text{ g/cm}^3 \Rightarrow F_g = 0 \text{ floats! } (\forall r)$$

$$\rho_{\text{iron}} \sim 7.8 \text{ g/cm}^3 \Rightarrow F_g = (7874 - 1000) \frac{\text{kg}}{\text{m}^3} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot \frac{4}{3} \pi (5\mu\text{m})^3$$

$$= 11 \text{ pN} \sim \frac{F_{\text{drag}}}{10}$$

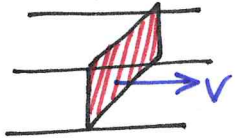
b) if the channel has a rectangular cross section,



$$w = 100\mu\text{m}$$

$$h = 20\mu\text{m}$$

what input flow rate is necessary
to obtain this velocity?

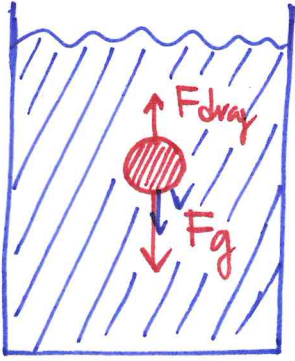


$$V = \frac{Q}{A} \rightarrow Q = V \cdot A$$

$$= \frac{1\mu\text{m}}{\text{ms}} \cdot 100\mu\text{m} \cdot 20\mu\text{m} = \frac{2 \cdot 10^{-15} \text{ m}^3}{\text{ms}} \cdot \frac{1000}{1000} = 2 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}}$$

$$= 2 \cdot 10^{-12} \cdot 10^9 \cdot 60 = \boxed{0.12 \mu\text{L/min}} \text{ (low flow rate)}$$

- (5) Find the terminal velocity of an iron sphere of $10\mu\text{m}$ diameter falling in water.



In steady state conditions:

$$\vec{F}_{\text{drag}} + \vec{F}_g = 0$$

$$F_{\text{drag}} = F_g$$

$$6\pi\eta r \cdot v = (\rho_p - \rho_{\text{fluid}}) \cdot g \cdot \frac{4}{3}\pi r^3$$

$$v = \frac{2}{9} \frac{(\rho_p - \rho_{\text{fluid}}) \cdot g \cdot r^2}{\eta}$$

$$= \frac{\frac{2}{9} (7874 - 1000) \cdot \frac{\text{kg}}{\text{m}^3} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot (5 \cdot 10^{-6})^2 \cdot \text{m}^2}{10^{-3} \text{ Pa} \cdot \text{s}} = \boxed{374 \mu\text{m/s}}$$

$$\frac{\frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{m}}{\text{s}^2} \cdot \text{m}^2}{\text{Pa} \cdot \text{s}}$$

$$= \frac{\text{kg}}{\text{s}^2} \frac{1}{\text{Pa} \cdot \text{s}}$$

$$= \frac{\text{kg}}{\text{s}^2} \frac{1}{\frac{\text{kg}}{\text{m} \cdot \text{s}}} = \left[\frac{\text{m}}{\text{s}} \right]$$

$$P_a = \frac{N}{\text{m}^2} = \frac{\text{kg} \cdot \text{m}}{\text{s}^2 \cdot \text{m}^2} = \frac{\text{kg}}{\text{m} \cdot \text{s}^2}$$

⑥ ELECTROOSMOSIS

Find the voltage that has to be applied to drive the flow in the following microchannel at a flow rate $Q = 1 \mu\text{L}/\text{min}$.

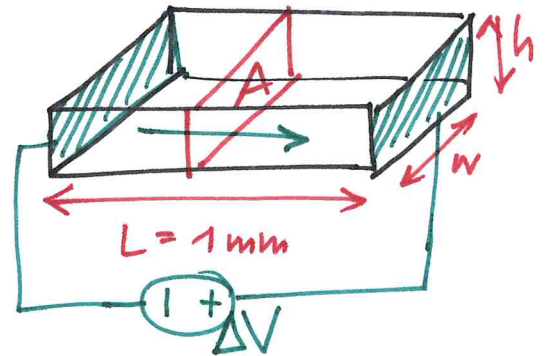
DATA: $\xi = 50 \text{ mV}$ (z-potential)

$$\eta_{\text{WATER}} = 10^{-3} \text{ Pa}\cdot\text{s}$$

$$\epsilon_{\text{WATER}} = 80 \cdot \epsilon_0$$

$$W = 100 \mu\text{m}$$

$$h = 10 \mu\text{m}$$



$$Q = v \cdot A = v \cdot (W \cdot h) \Rightarrow v = \frac{Q}{A} = \frac{1 \mu\text{L}/\text{min}}{W \cdot h}$$

$$= \frac{1 \cdot 10^{-6} \cdot 10^{-3} \frac{\text{m}^3}{\text{s}}}{60} = 0.016 \text{ m/s}$$

$$\frac{10 \mu\text{m} \cdot 100 \mu\text{m}}$$

$$V_{\text{ESM}} = \frac{\epsilon \cdot E \cdot \xi}{\eta} \Rightarrow \Delta V = E \cdot L = \frac{v \cdot \eta}{\epsilon \xi} \cdot L$$

$$\Delta V = \frac{0.016 \frac{\text{m}}{\text{s}} \cdot 10^{-3} \text{ Pa}\cdot\text{s}}{80 \cdot 8.86 \cdot 10^{-12} \frac{\text{F}}{\text{m}} \cdot 50 \text{ mV}} \cdot 1 \text{ mm} = \boxed{451 \text{ V}}$$

controlla dimensionale

$$\left[\begin{array}{l} \frac{\frac{\text{m}}{\text{s}} \cdot \text{Pa}\cdot\text{s}}{\frac{\text{F}}{\text{m}} \cdot \text{V}} \cdot \text{m} \\ \frac{\frac{\text{m}}{\text{s}} \cdot \frac{\text{N}}{\text{m}^2} \cdot \text{s}}{\text{J}/\text{m}} \end{array} \right] v = [\text{V}]$$

$$\frac{1}{2} c v^2 = \text{J}$$

$$\text{Nm} = \text{J} \quad \text{Pa} = \frac{\text{N}}{\text{m}^2}$$

if $L = 1 \text{ cm} \Rightarrow \Delta V \sim 4500 \text{ V}!!$

⑦ ELECTROPHORESIS

Consider the following system for DNA separation

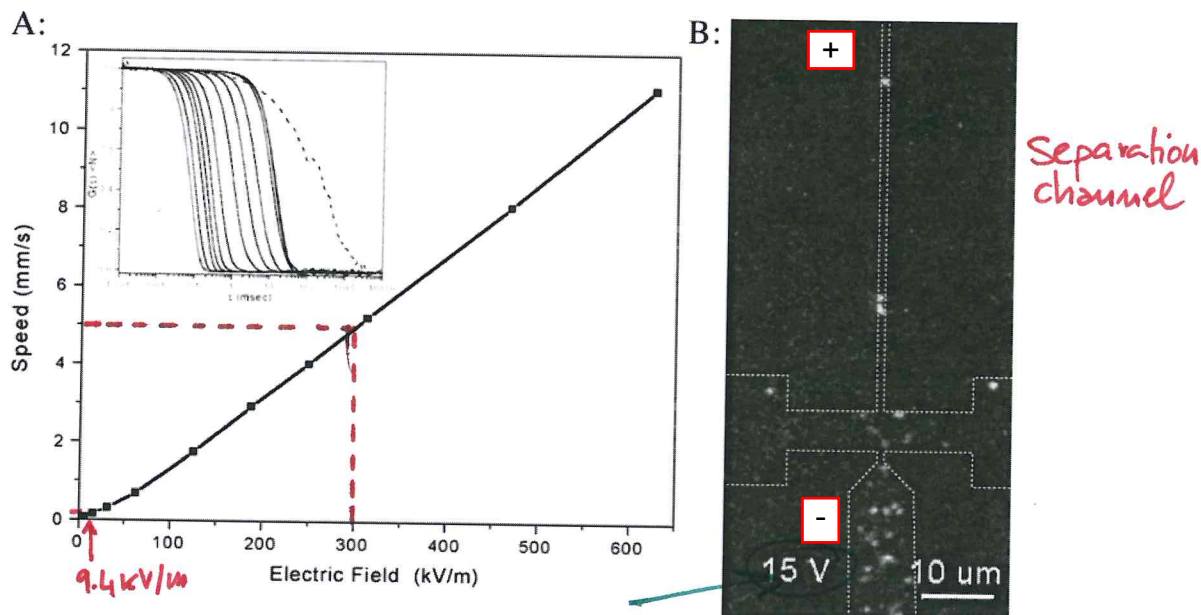


Figure 3. (A) Dependence of flow speed of DNA molecules in the microchannel on the applied electric field, using M13 dsDNA (7.3 kb). The inset shows the corresponding normalized autocorrelation curves at various voltages from which flow speeds were derived. The red curve is the fit to the FCS curve at 1000 V (625 kV/m), using a model assuming only one-dimensional translational flow (eq 2). The dashed FCS curve corresponds to conditions where the voltage was switched off. It is particularly noisy due to the rareness of single molecule event in the case of free diffusion (see text). (B) The image presents a picture taken with an Intensified CCD camera of a device under use. The outline of the capillary has been added for clarity. Time series of the motion of single molecules of DNA under the influence of an electric field can be downloaded at <http://www.hgc.cornell.edu/Microfluidics/Movies.html> (see text).

- What is the speed of the DNA molecule when 15V are applied?
- What is the charge of a DNA molecule?
- What is the frame rate of the CCD camera used?

a) If 1000V correspond to $E = 625 \text{ kV/m}$, $\Rightarrow$ the distance between the electrodes $L = \frac{\Delta V}{E} = \frac{1000 \text{ V}}{625 \text{ kV/m}} = 1.6 \text{ mm}$

when applying $\Delta V = 15 \text{ V} \Rightarrow E = \frac{15 \text{ V}}{1.6 \text{ mm}} = 9.37 \text{ kV/m}$

from the graph $\Rightarrow V = 0.15 \text{ mm/s}$

b) $V = \mu \cdot E$ $\mu = \frac{Q}{6\pi\eta r_g}$ r_g for DNA $\sim 1 \mu\text{m}$

from the graph $\rightarrow E = 300 \text{ kV/m} \rightarrow V = 5 \text{ mm/s}$

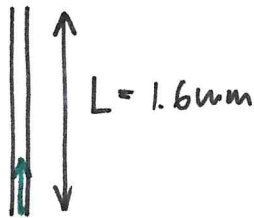
$\mu = \frac{V}{E} = \frac{5 \text{ mm/s}}{300 \text{ kV/m}} = 16.6 \cdot 10^{-9} \text{ m}^2/\text{Vs}$

$$Q = \mu \cdot (6\pi \eta r_g \cdot \eta) = 16.6 \cdot 10^{-9} \frac{\text{m}^2}{\text{Vs}} (6 \cdot \pi \cdot 1 \mu \cdot 10^{-3} \text{Pa} \cdot \text{s})$$

$$= \boxed{314 \cdot 10^{-18} \text{ C}} \quad (\# \text{ electrons } \frac{314 \cdot 10^{-18} \text{ C}}{1.6 \cdot 10^{-19} \text{ C}} \sim 2000 e^-)$$

Since the strand has 7.3 kb it is equivalent to $\sim$
 $\frac{2000 e^-}{7300 \text{ bp}} \sim 1 \text{ ion every 3 bases}$ (NEGATIVE CHARGE)

c) CCD:



$$T_{\text{travel}} = \frac{L}{V}$$

10 frames

$$\frac{1}{\text{frame rate}} \ll T_{\text{travel}}$$

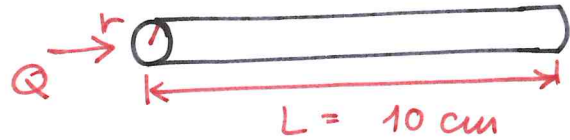
$$\text{Frame rate} > 10 \cdot \frac{1}{T_{\text{travel}}} = 10 \cdot \frac{V}{L} = 10 \cdot \frac{0.15 \text{ mm/s}}{1.6 \text{ mm}} = \boxed{1 \text{ frame/sec}}$$

↓
($T_{\text{travel}} \sim 10 \text{ s}$)

⑧ Considering the COMPRESSIBILITY of water, find the hydraulic time constant associated with a circular microchannel:

$$r = 10 \mu\text{m}$$

$$\beta = 5 \cdot 10^{-5} \text{ atm}^{-1}$$



COMPRESSIBILITY : $\beta = -\frac{1}{V} \cdot \frac{\partial V}{\partial P} \longrightarrow \text{HYDRAULIC CAPACITANCE}$

$$\tau = R \cdot C$$

$$R = \frac{8\eta \cdot L}{\pi r^4}$$

$$\beta = -\frac{1}{V} \cdot \frac{\Delta V}{\Delta P}$$

$$Q = C \cdot \frac{\Delta P}{\Delta t} \Rightarrow C = \frac{Q \cdot \Delta t}{\Delta P} = \frac{\Delta V}{\Delta P}$$

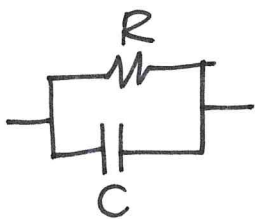
$$C = \frac{\Delta V}{\Delta P} = \beta \cdot V = \boxed{\beta \cdot \pi r^2 \cdot L}$$

$$\tau = \frac{8\eta \cdot L}{\pi r^4} \cdot \beta \cdot \pi r^2 \cdot L = \frac{8 \cdot \eta \cdot L^2}{r^2} \cdot \beta$$

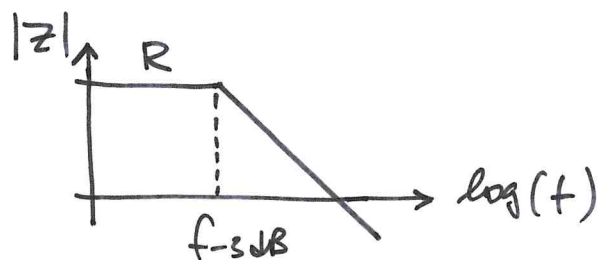
$$= \frac{8 \cdot 10^{-3} \text{ Pa s} (10 \cdot 10^{-2} \text{ m})^2}{(10 \cdot 10^{-6} \text{ m})^2} \cdot \frac{5 \cdot 10^{-5}}{\text{atm}} \quad [1 \text{ atm} = 10^5 \text{ Pa}]$$

$$= \frac{8 \cdot 10^{-3} (10 \cdot 10^{-2})^2}{(10 \cdot 10^{-6})^2} \cdot 5 \cdot 10^{-10} = \boxed{0.4 \text{ ms}}$$

$$f_{-3\text{dB}} = \frac{1}{2\pi\tau} = 400 \text{ Hz}$$

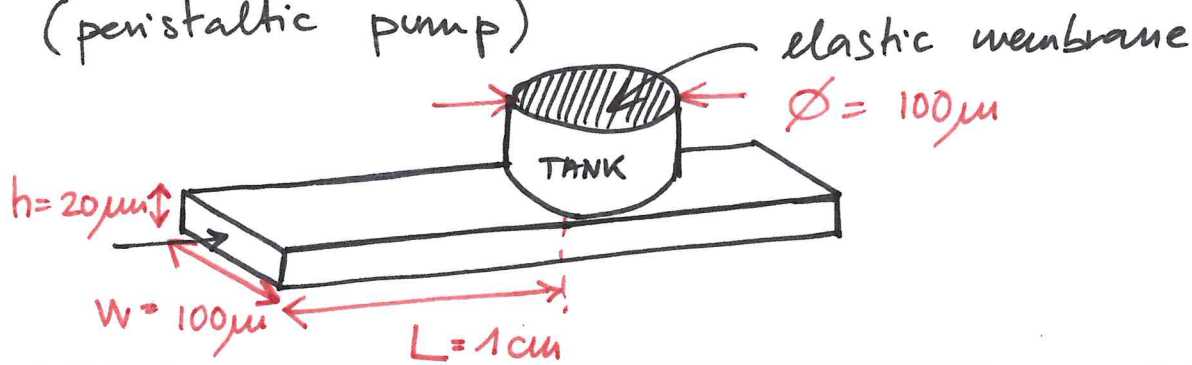


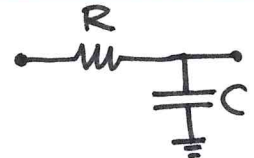
$$Z = \frac{R}{1 + s\tau}$$

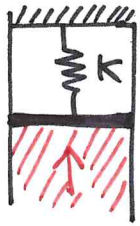


EQUIVALENT FLUIDIC IMPEDANCE $\longrightarrow$ LOW PASS FILTER

⑨ Considering the following microchannel connected to a tank, find the stiffness of the tank elastic membrane (spring constant) suitable to reduce by a factor 100 a periodic pulse of period $T = 2\text{ s}$ (peristaltic pump)



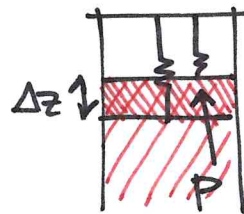
Equivalent model:  $\tau = R \cdot C$



the flexible membrane is modeled like a SPRING

$$C_e = \frac{\int i(t) dt}{V} \Rightarrow C_h = \frac{\int Q(t) dt}{P} = \frac{\Delta Vol}{P}$$

The tank volume expands:



$$\Delta Vol = Area \cdot \Delta z$$

From Hooke's law:

$$F = K \cdot \Delta z$$

$$F = P \cdot Area$$

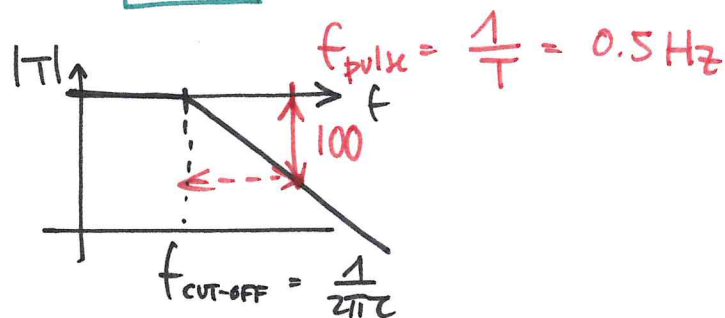
$$\Delta z = \frac{F}{K} = \frac{P \cdot A}{K}$$

$$\Delta Vol = A \cdot \Delta z = \frac{A \cdot P \cdot A}{K}$$

$$C = \frac{\Delta Vol}{P} = \frac{A^2 \cdot P}{K} \cdot \frac{1}{P} = \boxed{\frac{A^2}{K}}$$

$$T = \frac{1}{1 + 5\tau}$$

$$|T| =$$



for a 100-fold attenuation $\rightarrow f_{cut-off} = \frac{1}{100} \cdot f_{pulse} = \frac{1}{100} \cdot 0.5\text{ Hz}$

$$f_{cut-off} = 5\text{ mHz} \rightarrow \tau = \frac{1}{2\pi f_{cut-off}} = 31.8\text{ sec.}$$

$$\tau = R \cdot C = 31.8 \text{ s}$$

$$R = \frac{12 \eta L}{(1 - 0.63 \frac{h}{w}) h^3 w} \left(\frac{w}{h} = 5 > 1.4 \right)$$

$$= \frac{12 \cdot 10^{-3} \text{ Pa s} \cdot 10^{-2} \text{ m}}{(1 - 0.126) \cdot (20 \mu\text{m})^3 \cdot 100 \mu\text{m}} = 171.6 \cdot 10^{12} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3}$$

$$C = \frac{\tau}{R} = \frac{31.8 \text{ s}}{171.6 \cdot 10^{12} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3}} = 185.3 \cdot 10^{-15} \left[\frac{\text{m}^3}{\text{Pa}} \right]$$

$$C = \frac{A^2}{K} \rightarrow K = \frac{A^2}{C} = \frac{[\pi \cdot (50 \mu\text{m})^2]^2}{185.3 \cdot 10^{-15} \text{ m}^3/\text{Pa}} \left[\frac{\text{Pa} \cdot \text{m}}{\text{N/m}} \right]$$

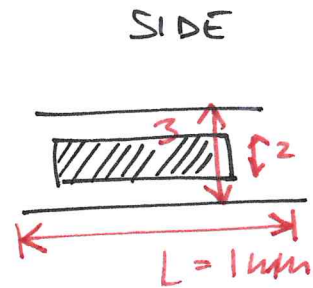
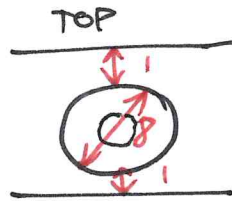
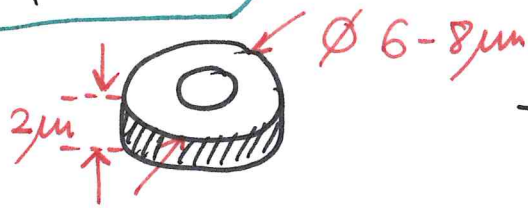
$$K = 0.33 \cdot 10^{-3} \text{ N/m}$$

If the spring constant $K = 0.1 \text{ N/m}$, find the tank diameter:

$$0.1 \frac{\text{N}}{\text{m}} = \frac{(\pi r^2)^2}{C} \quad \frac{\pi^2}{C} \cdot r^4 = K \quad r = \sqrt[4]{\frac{K \cdot C}{\pi^2}}$$

$$r = \sqrt[4]{\frac{0.1 \cdot 185.3 \cdot 10^{-15}}{\pi^2}} = 208 \mu\text{m} \rightarrow \varnothing 416 \mu\text{m}$$

⑩ Determine the dimension that a microfluidic chip would need to align and move red blood cells with a flow rate $Q = 1 \mu\text{e}/\text{min}$, driving the fluid by hydrodynamic forces.



$$R = \frac{12 \eta L}{(1 - 0.63 \frac{h}{w}) h^3 w}$$

should use $\eta_{\text{BLOOD}} > \eta_{\text{WATER}}$

$$h = 3 \mu\text{m}, w = 10 \mu\text{m} \quad \left(\frac{w}{h} > 1.4 \right) \quad \left| \begin{array}{l} \text{SMALLEST} \\ \text{CHANNEL} \\ \text{RISK OF} \\ \text{CLOGGING} \end{array} \right.$$

$$= \frac{12 \cdot 10^{-3} \text{Pa} \cdot \text{s} \cdot 1 \cdot 10^{-3} \text{m}}{(1 - 0.63 \cdot \frac{1}{3}) \cdot (3 \mu\text{m})^3 \cdot 10 \mu\text{m}} = 56 \cdot 10^{15} \frac{\text{Pa} \cdot \text{s}}{\text{m}^2}$$

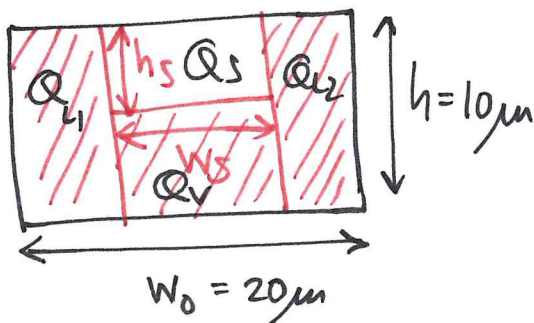
$$1 \text{e} = 10^{-3} \text{m}^3$$

$$\Delta P = Q \cdot R = 1 \mu\text{e}/\text{min} \cdot R = \frac{10^{-6}}{60} \frac{\text{e}}{\text{s}} \rightarrow \frac{10^{-9} \text{m}^3}{60} \cdot 56 \cdot 10^{15} =$$

$$= 937.64 \text{ kPa} \quad (93 \text{ atm})$$

TOO HIGH!
(6.8 atm max syringe pump)

SOLUTION!: 3D hydrodynamic FOCUSING



$$h_s = h_0 \cdot \frac{Q_s}{Q_s + Q_v} = 3 \mu\text{m}$$

$$w_s = w_0 \cdot \frac{Q_s + Q_v}{Q_s + Q_v + Q_L} = 4 \mu\text{m}$$

$$Q_s = 1 \mu\text{e}/\text{min} \rightarrow Q_v = 3 \mu\text{e}/\text{min} \quad (h_s = 10 \cdot \frac{1}{1+3} = 2.5 \mu\text{m})$$

$$Q_{L1} = Q_{L2} = 5 \mu\text{e}/\text{min} \rightarrow Q_L = 10 \mu\text{e}/\text{min} \quad (w_s = 20 \cdot \frac{4}{4+10} = 6.8 \mu\text{m})$$

$$Q_{\text{TOT}} = Q_s + Q_v + Q_L = 1 + 3 + 10 = 14 \mu\text{e}/\text{min}$$

||
2Q_{L1}

$$R = \frac{12 \eta L}{(1 - 0.63 \frac{h}{w}) h^3 w} = \frac{12 \cdot 10^{-3} \text{ Pa} \cdot 10^{-3} \text{ m}}{(1 - 0.63 \frac{1}{2}) \cdot (10 \mu\text{m})^3 \cdot 20 \mu\text{m}} = 875.9 \cdot 10^{12} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3}$$

$$\Delta P = 14 \mu\text{L/min} \cdot R = \frac{14 \cdot 10^{-9}}{60} \frac{\text{m}^3}{\text{s}} \cdot 875.9 \cdot 10^{12} = 204 \text{ kPa}$$

(20 atm)
↓

still too much

(I have to reduce Q_s to $0.5 \mu\text{L/min}$
and half all the flow-rates to use a syringe pump)

or use 4 syringes

